

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION: SALEM 1 & 2

SYSTEM: Emergency Operating Procedures

TASK: Implement actions required for a Natural Circulation Cooldown

TASK NUMBER: N1150530502

JPM NUMBER: 13-01 NRC Sim e

ALTERNATE PATH: ☒

K/A NUMBER: EPE E09 EA1.1

IMPORTANCE FACTOR: 3.5 3.5

APPLICABILITY:

EO ☐

RO ☒

STA ☐

SRO ☒

RO

SRO

EVALUATION SETTING/METHOD: Simulator - Perform

REFERENCES: 2-EOP-TRIP-4, Natural Circulation Cooldown, Rev. 23

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 30 minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By: G Gauding
Instructor

Date: 9-6-14

Validated By: C Omlor
SME or Instructor

Date: 10-15-14

Approved By: *[Signature]*
Training Department

Date: 10-23-14

Approved By: *[Signature]*
Operations Department

Date: 10-23-14

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:

GRADE: ☐ SAT ☐ UNSAT

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE:

DATE:

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TASK: Implement actions required for a Natural Circulation Cooldown

TASK NUMBER: 1150050501

Simulator Setup:

IC-255

MALF EL0134 Loss of Offsite Power

MALF RP0108 Failure of Automatic SI

Overrides B106-B108 to override 2CV3,4,5 OPEN PB's OFF, and CLOSE PBs ON ties to Event-1 which is 2CV18 AUTO PB, which will isolate letdown and prevent is being put back in service.

Overrides B201-202 which prevents closing 2PR1, 2PR2, 2PR6, and 2PR7.

SIMULATOR OPERATOR: Open Data Collection program and trend all 4 RCS loop Tcs, then print trends when JPM is complete.

INITIAL CONDITIONS:

Unit 2 tripped from 100% power when a loss of all off-site power occurred. All vital buses loaded in MODE 2 SEC operation. Operators performed TRIP-1 and TRIP-2, then transitioned to TRIP-4 to perform a Natural Circulation Cooldown. Operators are performing Step 9 to establish a 20-25°F/hr cooldown rate using 21-24MS10s. Current RCS Tc C/D rate is 5°F/hr. Letdown was unable to be placed in service when directed in TRIP-2. Control Console bezels have been swapped out, and Letdown may be placed in service when directed by procedure.

INITIATING CUE:

You are directed to raise the C/D rate to 20-25°F/hr using 21-24MS10 as directed at step 9.1 of TRIP-4, then continue with remainder of TRIP-4.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion

1. Raise C/D rate to 20-25°F per hour.
2. Depressurize the RCS by opening ONE PORV.
3. Initiate Safety Injection after PORV fails to close prior to the loss of RCS subcooling (~ 3 minutes after PORV is opened, based on CT Book, Appendix B.2, page 2, italicized text)

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TASK: Raise C/D rate and Lower RCS Pressure during a natural circulation cooldown IAW EOP-TRIP-4

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide candidate with the "Tear Off Sheet"	Reviews the conditions and marked up EOP		
		START TIME:			
	9.1	CONTROL STEAM FLOW TO MINIMIZE SG DELTA-P			
*		DUMP STEAM USING <u>INTACT</u> SG MS10S (RELIEF VALVES)	Adjusts 21-24MS10s to raise cooldown rate while minimizing SG Delta-P.		
		MAINTAIN SG NR LEVEL BETWEEN 9-33%	Maintains SG NR level between 9-33% by adjusting 21-24AF21 valves		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	10	WAIT UNTIL ALL RCS THOTS ARE LESS THAN 543°F	Monitors RCS Thots while raising cooldown rate. After candidate has modified C/D, then Cue: The CRS directs you to report the current RCS Tc Cooldown rate. Evaluator monitor candidate actions in response to the cooldown rate provided to CRS, and actions taken in regard to what the cooldown rate is, if cooldown rate is determined to be too high/low, candidate adjust MS10 position to raise or lower cooldown rate. Once this is performed, provide the next cue as directed. Cue when RCS Tavg is <543°F: "All RCS Thots are now <543°F. Another operator will now monitor and control the RCS cooldown while you continue in TRIP-4.		
*	11	BLOCK HI STEAM LINE FLOW SI	Blocks HI STEAM LINE FLOW SI on both trains		
	12	OPEN CV2 AND CV277 (LETDOWN CONTROL VALVES) AND PLACE IN "AUTO"	Ensures open CV2 and CV277 (LETDOWN CONTROL VALVES) and selects AUTO for each valve		
		OPEN CV7 (LETDOWN CONTROL VALVE)	Opens CV7(LETDOWN CONTROL VALVE)		

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SYSTEM: **Emergency Operating Procedures**

TASK: **Raise C/D rate and Lower RCS Pressure during a natural circulation cooldown IAW EOP-TRIP-4**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		ADJUST CV55 (CHARGING FLOW CONTROL VALVE) TO RAISE CHARGING FLOW TO <u>AT LEAST</u> 87 GPM	Adjusts CV55 (CHARGING FLOW CONTROL VALVE) to obtain no less than 87 gpm on the charging flow indicator.		
		PERFORM THE FOLLOWING ACTIONS SIMULTANEOUSLY: • OPEN <u>ONLY ONE</u> ORIFICE ISOLATION VALVE • ADJUST CV18 (LETDOWN PRESSURE CONTROL VALVE) TO MAINTAIN LETDOWN PRESSURE AT 300 PSIG	Cue if required: If requested, direct the operator to place a 75 gpm orifice in service (CV4 or 5). Opens only one (Letdown Orifice Isolation Valves) and adjusts CV18 to control letdown pressure such that the letdown line relief valve does not lift (600 psig).		
*		PLACE THE FOLLOWING IN AUTO: • CV18 • MASTER FLOW CONTROLLER • CV55	• CV18 in AUTO* • Master Flow Controller in AUTO • CV55 in AUTO * When the 2CV18 is placed in auto, it will shut, along with any open orifice valve, and all orifice valves will be prevented from opening.		
	13	IS LETDOWN IN SERVICE	Answers NO		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	13.1	<u>CAUTION:</u> PZR PORV CYCLING SHOULD BE MINIMIZED	Reads statement		
*		USE <u>ONLY ONE</u> PZR PORV TO LOWER RCS PRESSURE TO 1865 PSIG	Opens only 2PR1 <u>OR</u> 2PR2		
		WAIT <u>UNTIL</u> RCS PRESSURE LESS THAN 1865	Monitors RCS pressure		
*		CLOSE <u>BOTH</u> PZR PORV's	Attempts to shut the open PORV, and determines it will not shut. Attempts to shut associated PORV Block valve, and determines it will not shut. Initiates Safety Injection based upon impending CAS action of loss of subcooling or automatic safety injection nearing/reaching setpoint. (Auto SI actuation is failed in sim setup.)		
			Terminate JPM when SI has been initiated.		
		STOP TIME:			

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- 1 1. Task description and number, JPM description and number are identified.
- 2 2. Knowledge and Abilities (K/A) references are included.
- 3 3. Performance location specified. (in-plant, control room, or simulator)
- 4 4. Initial setup conditions are identified.
- 5 5. Initiating and terminating Cues are properly identified.
- 6 6. Task standards identified and verified by SME review.
- 7 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 8 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev 23 Date 10-15-14
- 9 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: OMLOK

Date: 10-15-14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

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